IKE NANCY NNENNA
VUG/SEN/24/12359

PART 1

ENTITY	ATTRIBUTES	PRIMARY KEY
Department	Department_id, Department_name	Department_id
Student	Matric_number, Department, Name	Matric_number
Lecturer	Lecturer_Id, Name	Lecturer_Id
Course	Course_code, Course_title, Credit_unit, Credit_load	Course_code
Entrollment	Matric_number, Course_code, session	(student_id, course_id, session)

STEP 2

i. One student belongs to one department.
One department can have many students
Relationship: One to Many (1:M)

ii. How many courses can one student register for? How many students can one course have? A
One student can register for many courses
One course can have many students
Relationship: Many to Many (M:M)

iii. How many lecturers can one course have across sessions? How many courses can one lecturer teach?
One course can have many lecturers
One Lecturer can teach many courses
Relationship: Many to Many (M:M)

PART 2

matric_no	student_name	student_dept	course_codes	course_titles	lecturers	lecturer_depts	session
20231234	Ada Obi	Software Engineering	SEN307, SEN301	Database Systems, Software Architecture	Dr A. Okafor, Dr M. Bello	Software Eng., Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307, COS301	Database			

Systems,
Operating
Systems Dr A.
Okafor,
Mrs F. Eze Software Eng.,
Computer Sci. 2026/2027
20231236 Grace Okon Software
Engineering SEN301 Software
Architecture Dr M.
Bello Software Eng. 2026/2027

1st NORMALISATION

matric_no student_name student_dept course_codes course_titles
lecturers lecturer_depts session
20231234 Ada Obi Software
Engineering SEN307,
Database
Systems,
Dr A.
Okafor,
Software Eng.,
2026/2027
20231234 Ada Obi Software
Engineering SEN301 Software
Architecture Dr M.
Bello Software Eng. 2026/2027
20231235 John Musa
Computer
Science SEN307, Database
Systems,
Dr A.
Okafor,
Software Eng.,
2026/2027
20231235 John Musa Computer
Science COS301 Operating
Systems Mrs F. Eze Computer Sci. 2026/2027
20231236 Grace Okon Software
Engineering SEN301 Software
Architecture Dr M.
Bello Software Eng. 2026/2027

Primary key: (matric_no, course_code, session)

Explanation: I removed the multi-valued cells and gave each student–course pair its own row, making every cell contain one value.

2nd NORMALISATION

STUDENT TABLE

MATRIC NUMBER	STUDENT_NAME	STUDENT_DEPARTMENT
20231234	Ada Obi	Software Engineering
20231235	John Musa	Computer Science
20231236	Grace Okon	Software Engineering

COURSE

COURSE CODES	COURSE TITLES
SEN307	Database Systems,
SEN301	Software Architecture
COS301	Operating Systems

ENROLMENT

Matric_No	Course_code	Session
20231234	SEN307	2026/2027
20231234	SEN301	2026/2027
20231235	SEN307	2026/2027
20231235	COS301	2026/2027
20231236	SEN301	2026/2027

Explanation: I separated student and course details into their own tables so that non-key attributes depend on the whole primary key of their respective tables.

3RD NORMALISATION

DEPARTMENT

DEPT_ID	Dept_Name
DO1	Software Engineering
DO2	Computer Science

STUDENT

matric_no (PK)	student_name	dept_id (FK)
20231234	Ada Obi	DO1
20231235	John Musa	D02
20231236	Grace Okon	D01

Explanation: I separated student and course details into their own tables so that non-key attributes depend on the whole primary key of their respective tables.

COURSE
COURSE CODES COURSE TITLES
SEN307 Database
Systems,
SEN301 Software
Architecture
COS301 Operating Systems

LECTURER

lecturer_id (PK) lecturer_name dept_id (FK)
L01 Dr A. Okafor D01
L02 Dr M. Bello D01
L03 Mrs F. Eze D02

ENROLMENT
Matric_No Course_code Session
20231234 SEN307 2026/2027
20231234 SEN301 2026/2027
20231235 SEN307 2026/2027
20231235 COS301 2026/2027
20231236 SEN301 2026/2027

COURSE LECTURER
lecturer_id (PK, FK) course_code (PK, FK) session (PK)
L01 SEN307 2026/2027
L02 SEN301 2026/2027
L03 COS301 2026/2027

Explanation: I moved department details into a Department table and lecturer details into a Lecturer table, linking them with foreign keys to eliminate transitive dependencies and reduce redundancy.

```
Department (
    dept_id PK,
    dept_name
)

Student (
    matric_no PK,
    student_name,
    dept_id FK → Department.dept_id
)
```

```
Lecturer (
    lecturer_id PK,
    lecturer_name,
    dept_id FK → Department.dept_id
)

Course (
    course_code PK,
    course_title
)

Enrolment (
    matric_no PK, FK → Student.matric_no,
    course_code PK, FK → Course.course_code,
    session PK
)

Course_Lecturer (
    lecturer_id PK, FK → Lecturer.lecturer_id,
    course_code PK, FK → Course.course_code,
    session PK
)
```